

# ATM over copper

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*ATM is often associated with high-speed, long-distance communications and therefore the use of fibre optic cable as the transmission medium. However, when the advantages which ATM can bring are considered, there is a strong case for its use in other environments which may favour different, in particular copper-based, transmission media. This paper reviews the rationale and status of various activities which aim to specify and promote the transmission of ATM over copper.*

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## 1. Introduction

The main strength of ATM lies in its ability to efficiently transport a variety of different traffic with a well-defined quality of service over a common transmission link. This has two major benefits:

- it permits traffic aggregation and a consequent statistical multiplexing gain, e.g. for PBX interconnection,
- it allows the controlled delivery of a mixture of traffic types (e.g. voice, video, data) to a single end point, as is required for true multimedia services.

The former case is associated with long distance, wide area or public network connections in which case fibre optic transmission is usually the most efficient way of meeting the bandwidth/distance requirements. However, in the latter case, connection is required to devices which may well already be attached to the infrastructure of a private network — this infrastructure is commonly known as building wiring. The vast majority of installed building wiring is copper based; there is very little fibre to the desk. Much of this wiring has been put in over the past three years as the advantages of structured cabling have become apparent and cabling standards have matured. Wiring is seen as a long term (10—15 year) investment by many companies; this is due not only to the high capital cost, but the major disruption which installing a new cabling scheme can produce. There is a natural reluctance to install a new infrastructure simply to be able to have access to a new service — one of the drivers behind the development of

structured cabling was that new services should be designed around existing infrastructure types and configurations.

The widespread adoption of ATM for the delivery of multimedia services therefore depends upon being able to deliver ATM to the desktop over existing copper infrastructure types. This paper discusses the issues involved with achieving this, and reports on various relevant specification activities.

Complete end-to-end ATM systems are still a rarity in the business environment due to the immaturity and high cost of the technology. The idea that ATM might be appropriate in the residential environment would have been regarded as extremely far fetched until fairly recently; however, there is now a great deal of interest in this topic, and ironically this may be the first area which sees the widespread deployment of ATM. Recent developments in residential ATM are therefore also discussed.

## 2. A building wiring primer

While this paper attempts to give sufficient information to enable the main arguments to be understood by a general reader, it has not been possible to explain every concept in detail in the space available. However, it is particularly important to have some understanding of the terminology and rationale regarding building wiring in general and structured cabling in particular (the Appendix contains a listing of commonly used acronyms). Private, in-building data networks were originally used for the connection of

mainframe computers to fairly dumb terminals. These systems were completely proprietary, including the wiring which was usually some kind of co-axial or twin-axial cable. Such cables were expensive, difficult to install and modify, and only suitable for one particular type of system. With the move to more distributed and open computing based on PCs interconnected by LANs, this approach to cabling became unacceptable. Users wanted to be able to migrate to different, more powerful systems, possibly involving a LAN having a different topology, without having to change the basic infrastructure. Staff mobility also became more common, so convenient access to a widespread, general-purpose, ubiquitous infrastructure was important.

The requirements for what became known as structured cabling were therefore as follows:

- low cost,
- simple,
- physically compact and flexible,
- non-proprietary (cable and connector type),
- provide support for different systems,
- adaptable topology,
- ubiquitous within a building.

These aims were realised as follows, and illustrated in Fig 1. Unscreened, twisted pair (UTP) cable with RJ45 connectors were deemed to be the most suitable basic components. 4-pair cables radiate from a central wiring closet to data-jack units (DJUs) arranged on a regular grid pattern, into which user equipment can be plugged by means of patch-cords. The cheapness of the system allows over-provisioning, i.e. more outlets than users, which provides the required positional flexibility. The physical star can be configured as a star, ring or bus by simply repatching at the central location, thus supporting all the common LAN topologies. The maximum distance from the wiring closet to any DJU is normally limited to 100 m. Surveys have shown that it is possible to position a wiring closet such that it is no further than 100 m from the most distant point on the same floor of a building in the overwhelming majority of cases. This distance is now used as the design reach for most new LAN systems.

It is interesting to note that UTP cable was chosen for its logistical qualities (cost, flexibility, ease of installation) rather than its transmission properties; there are other cable types which are superior in that respect but which do not meet the logistical needs.

It is not sufficient to simply state the cable type, the transmission properties also need to be specified to enable

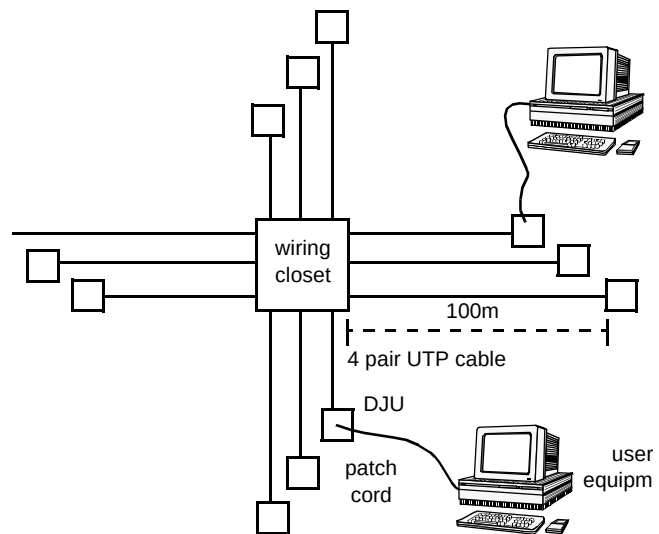


Fig 1 Typical structured cabling layout.

system designers to produce suitable equipment. This specification has now been done initially by the American EIA [1] and latterly by ISO [2], according to a 'category' system. The lowest quality UTP cable type used for data systems today is category 3 — this is specified up to 16 MHz and has traditionally been used to support systems such as Ethernet and Token Ring. The vast majority of new installations use the superior category 5 cable which is specified up to 100 MHz. The primary difference between the two cable types is their crosstalk and attenuation performance, with the superiority of category 5 UTP being due to both its constitution (a lower loss dielectric) and manufacture (tighter and better controlled twists). Category 4 is an intermediate cable type which has now largely fallen into disuse.

### 3. Copper developments and capabilities

In order for the concept of ATM delivery over the existing cable types described above to be useful, these cables must be able to support the kind of bandwidths that need to be delivered. This raises an immediate problem in that the services and therefore bandwidths have not yet been defined; however, certain general ground rules can be applied. ATM needs to be able to offer both an enhanced functionality and bandwidth compared to the most common LAN protocols (Ethernet and Token Ring) in order to persuade users to invest in new equipment. Further, part of the reason behind the drive towards ATM is the desire to be able to deliver video which remains fairly bandwidth hungry, even given modern compression techniques. Therefore a bandwidth at least several times that of 16 Mbit/s Token Ring is indicated.

There is an existence of proof that this can be achieved. A standard (ANSI) exists for a copper version of FDDI [3]

which has a line rate of 125 Mbit/s (100 Mbit/s data rate with 4B5B encoding) and can operate over 100 m of category 5 cable. Standards-compliant equipment exists and is in fairly widespread use. Therefore at the time that ATM over copper development started, there was already a good indication that the idea was feasible.

It is interesting to contrast this development with the market expectation of a few years ago. Ethernet and Token Ring systems were operating over category 3 cable, but the view was that significantly higher bandwidths could only be achieved on fibre. Although no-one had a requirement for very high bandwidths at that time, there was a desire to ensure that newly installed infrastructures had some upgrade potential, but without the expense of installing fibre before it was needed. This led to the development of blown fibre which enables a cheap plastic tube to be installed (by unskilled labour) alongside conventional copper cabling and into which fibre bundles can be blown using compressed air at a later date, if required.

What in fact happened was that copper developments have more than kept pace with increased bandwidth demands, which has delayed the adoption of fibre (or even blown fibre tubing) to the desk indefinitely. These developments have been in both the cables themselves and the transmission technologies. The major cable development was the production of category 5. It is not a trivial manufacturing process to produce this cable but the biggest challenge was to upgrade the connector crosstalk performance so that it did not significantly degrade the cable. The strong market desire to stick with the current RJ45 connector type is the main reason that 'category 6' (i.e. a performance significantly better than category 5) systems are unlikely to be developed — category 6 cable would be possible, but category 6 RJ45 connectors nearly impossible to manufacture.

Given that the practical limit of this cable/connector combination has now been reached, further advances have depended on developments in transmission technology. The fact of stable international cable standards [1, 2] has been a very important factor in persuading silicon manufacturers that it is worth investing in more complex transmission technology. It should be noted, however, that there is a constraint here which places an upper limit on the complexity of such technology. There is no point making transmission equipment so complex and expensive that it would be cheaper (at a system level) to use a fibre optic based system.

#### 4. Problems peculiar to UTP cable

The use of UTP cable is now so widespread and effective that it is very easy to become blasé about it, and forget what had to be done to achieve high-speed transmission. In

order to explain this, it is necessary to describe in more detail what UTP cable is, and the factors which limit its performance.

UTP cable consists of one or more pairs of conductors twisted together in a controlled fashion (Fig 2). There are normally four pairs within a bundle which is contained within an outer flexible plastic sheath.

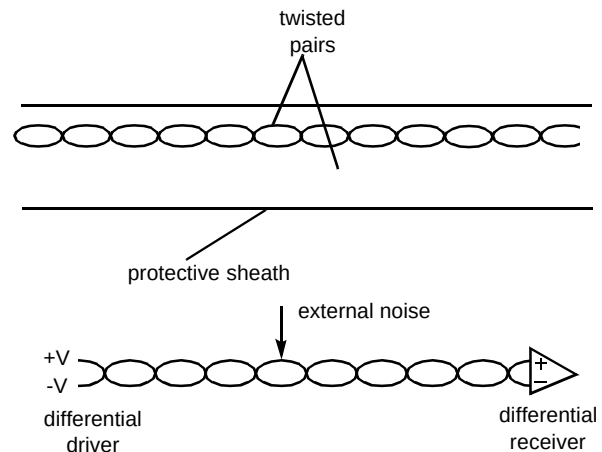


Fig 2 Twisted pair cable construction and operation.

This method of construction provides a cable with a reasonably well-defined characteristic impedance, and which has good emissions and noise-interference immunity properties when it is used as part of a balanced system. The two conductors of a pair are driven with equal and opposite signals so that the radiation arising from the individual wires cancels out. Similarly, any external noise should couple equally into both conductors and can therefore be removed by the use of a differential receiver. Systems are solely dependent upon the degree to which this balancing mechanism works to provide the required emissions and immunity performance — there is no screening either between the cable and the outside world, or between the individual pairs of a cable. It is the latter fact which gives rise to one of the performance limitations of UTP cable. The balance of the cable is not perfect and a certain proportion of the signal will be coupled across to adjacent pairs in the same (four-pair) bundle — these are physically very close. The amount of coupling is characterised by a parameter called the crosstalk attenuation. The amount of coupling increases with frequency by about 15 dB per decade, although the precise form of the coupling is a very complex function of frequency as various signal components add in and out of phase; the number of twists per unit length for each pair within a bundle is made different so as to minimise the interaction between the different pairs. A typical measured crosstalk coupling function is shown in Fig 3.

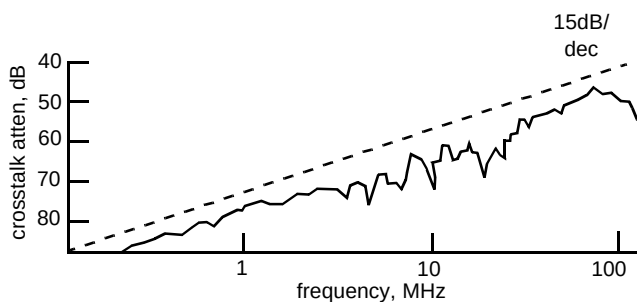


Fig 3 Typical measured crosstalk attenuation function for Category 5 UTP.

Many transmission links need to be full duplex, i.e. allow simultaneous transmission in both directions. For the baseband systems commonly used in LANs, it is not possible to combine the two directions on a single path by means of frequency separation; this is why the standard cable has more than one pair to allow duplex operation via simultaneous, simplex operation on 2 pairs. Consider the situation in Fig 4 which shows a 2-pair, full-duplex transmission link. The signal received at A is attenuated by the cable, but also subject to the crosstalk interference described above. Since both the cable attenuation and crosstalk coupling increase with frequency, at some point the crosstalk interference will exceed the real attenuated signal. This places a limit on the usable bandwidth of the cable which cannot be exceeded without going to a significantly more complicated transmission system.

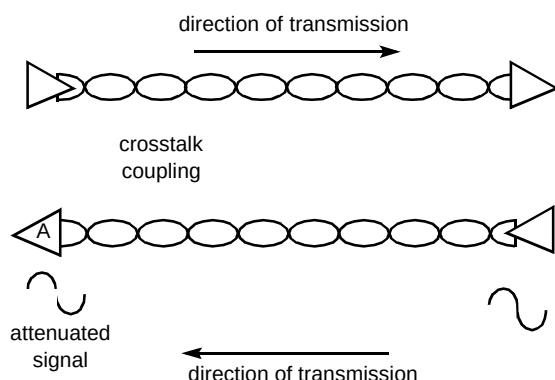


Fig 4 Near-end crosstalk interfering with the received signal.

Given the frequency dependence of crosstalk and attenuation, it is clearly desirable to limit the high frequency content of the signals, but this has to be done against the background of an ever-increasing desire for bandwidth. This can be achieved by the use of line codes which are more bandwidth efficient (in terms of bit/s/Hz) than simple binary codes. This means going to multi-level codes, such as the ternary MLT-3 used for CDDI, or multi-dimensional (amplitude and phase) codes such as CAP described in more detail in a later section. However, the cost of this approach is the increased complexity needed, particularly in the

receivers, although this has become more feasible with the advent of DSP-based adaptive equalisers.

Although this helps, there is a cost/complexity trade-off and as bandwidth demands continue to increase, crosstalk will again become a problem at some point. An alternative approach is therefore to use a relatively simple line code but employ a technique known as crosstalk cancellation; this uses the fact that the form of the interfering signal is known, to largely cancel out the crosstalk in an adaptive fashion. This method has again been made possible by advances in high-speed, cost-effective DSP; although it has not yet been widely deployed, it has been demonstrated to work in a real system.

Once a system contains a crosstalk canceller, this can be used for another purpose which offers the prospect of a further increase in bandwidth. A crosstalk canceller determines and applies a compensating signal to a receiver operating on a different pair to the transmitter. However, essentially the same technique can be used when the transmitter and receiver are on the same pair, in which case near-end echo can be cancelled, and full-duplex working achieved on a single pair. The bandwidth increase is then achieved by 'multi-pair' working, i.e. using each of the four pairs in full-duplex mode as opposed to two in simplex mode. A lower degree of multi-pair working — four simplex systems — can of course be employed without the need for echo cancellation, but this will not give as much bandwidth as full, four-pair, full-duplex operation.

So far the effects of imperfect balance between the pairs within a cable have been considered, but it is also necessary to worry about non-perfect balance with respect to the outside world in order to achieve the required emissions and immunity performance. To a certain extent this has been catered for as systems which constrain the bandwidth for transmission reasons will also have advantageous EMC properties. A cable will radiate when unbalanced or common mode currents flow. There are various mechanisms in the cable whereby a perfectly balanced, (launched) differential signal will be partially converted to a common mode signal during passage down the cable. This conversion increases with frequency. There is also an advantage from the regulatory point of view in constraining as much of the bandwidth as possible to below 30 MHz, as this is the point at which the radiated emissions regulations start [4].

Given the desire to again restrict the high-frequency content of signals, it is important to avoid strong spectral lines which have high-frequency harmonics. Such lines arise from repeating data patterns; this is particularly important in systems such as ATM which have the concept of a cell stream which is only partially filled by real data. If the unused cells contain a fixed idle pattern, then EMC problems could arise. However, nearly all new high-speed

systems (including ATM) now use (feedback shift register) scrambling techniques to randomise the data and avoid spectral line problems.

While there are technical limitations with UTP cable as a transmission medium, there are well-understood techniques whereby its capacity has been enhanced to a point well in excess of what was thought practical just a few years ago, and eminently suited to the transmission of high-speed systems such as ATM. UTP became popular because of its logistical rather than technical advantages, and was chosen in preference to an existing fully shielded cable type (IBM Type 1) which is technically superior. It is therefore sensible to concentrate on UTP as the cable type on which to focus the development of ATM over copper. Systems which have been defined for ATM over UTP copper are described in the following sections.

## 5. Business systems

All the initial activity to specify ATM over copper solutions was aimed at the private network business market. The residential development is a very recent activity, much less advanced and will be dealt with separately.

### 5.1 Requirements

There are three drivers for the business market from which detailed system requirements might be derived:

- what bandwidth existing (UTP) infrastructure types can support,
- what bandwidth the services require,
- what competing systems can offer.

Unfortunately the first two of these are not well defined. The desire to allow desktop connection over existing UTP cable types has been the major drive behind this activity, but as discussed above, the bandwidth support offered depends on the complexity of the systems in which one is prepared to invest. The services and therefore their bandwidth requirements are even less well known, although this whole activity only makes sense if real time (albeit compressed) video delivery can be offered and a 'high bandwidth' capability is therefore indicated. The third factor, however, does give more of a precise pointer. While ATM delivery offers many advantages for the delivery of a mixture of services, this is unlikely to be sufficient to persuade people to invest in new technology unless it also offers a higher bandwidth capability than their present system — how much bandwidth they actually use or need is quite immaterial. Therefore any ATM-based system must certainly exceed the capacity of current day systems such as 10 Mbit/s Ethernet and 16 Mbit/s Token Ring. There is also emerging competition from CDDI and Fast Ethernet which

both offer raw bit rates of 100 Mbit/s, and although effective bandwidth per user should be the main criterion, being at least comparable in raw, peak bit rate terms is seen to be important.

There was a desire to support both the majority installed base of category 3 UTP, and the superior category 5 cable used in all new installations. It would have been possible to specify a single set of requirements and a single solution for these, but this would have resulted in a compromise in some way, i.e. a high bandwidth solution which was unnecessarily complex for the higher quality cable, or a simpler solution which did not use the higher bandwidth capability of the better cable type. It was therefore decided to define two sets of requirements, and develop two different solutions, one for each cable type. These design aims are summarised in Table 1.

Table 1 System design aims.

|                 | Category 3 UTP              | Category 5 UTP              |
|-----------------|-----------------------------|-----------------------------|
| Bit rate        | 51.84 Mbit/s                | 155.52 Mbit/s               |
| Framing         | SONET                       | SONET                       |
| Reach           | 100 m                       | 100 m                       |
| EMC performance | Class A, Level 2<br>(3 V/m) | Class A, Level 2<br>(3 V/m) |

### 5.2 Current specifications

The ATM Forum is the prime mover in producing specifications for ATM-based systems, including ATM over copper. The Forum has completed its initial work in this area which was to find a solution for each of the sets of design criteria given in Table 1. This has resulted in two specifications, both now published — the mid-range PHY, for Category 3 UTP [5], and an NRZ-based, 155.52 Mbit/s system for Category 5 UTP [6]. These solutions will now be briefly described.

#### 5.2.1 Mid-range PHY for category 3 UTP

The mid-range (i.e. between low-speed rates such as E1 and the bottom of the SDH range, STM-1) PHY specifies a SONET framed, 51.84 Mbit/s system which uses 16-CAP encoding. CAP is a technique which utilises both amplitude and phase modulation and is essentially a variation of the well-known QAM method, but uses Hilbert filtering instead of explicit carrier modulation. The '16' refers to the number of points in the signal constellation. One of the attractions of this system is that the spectrum is well defined, and in this specification constrained to lie below 30 MHz, thus avoiding EMC emission problems. It is also possible to run the same system at one half or one quarter of the 51.84 Mbit/s rate by simply using only 4 or 2 of the constellation

points respectively (4-CAP or 2-CAP); the spectral properties remain exactly the same. These lower rates allow greater (than 100m) reach, or sub-category 3 cable to be used, both of which increase the proportion of the installed cable base which can be used.

There was a great debate within the ATM Forum about whether 51.84 or 25.92 Mbit/s was the correct rate for this mid-range solution, and whether the complexity of CAP was required. The two arguments are related; alternative line codes, such as BPRIV, were suggested, but, while simpler, and practical at 25 Mbit/s, would probably have needed crosstalk cancellation at 51 Mbit/s. The concern over complexity was that although CAP technology has been in use for some time, it has never been implemented at such high rates before.

### 5.2.2 155 Mbit/s over twisted pair cable

The title of this specification is deliberately less precise; although it was mainly targeted at Category 5 UTP, it also takes account of the existence of 150  $\Omega$  STP (IBM Type 1 cable) — parameters are derived for the (100  $\Omega$ ) UTP case and modified to match the 150  $\Omega$  cable where necessary. It operates at the STS-3 rate (155.52 Mbit/s) and uses SONET framing. The line code chosen is the simple binary NRZ. This choice is curious in that it has a very broad spectrum and it is not obvious that EMC emission requirements can be met. Indeed, the ANSI Committee developing the CDDI specification spent a long time coming to the conclusion that at least a three-level code was required from the EMC point of view, and that was for the lower line rate of 125 Mbit/s. The matter is complicated by the fact that although a certain level of EMC emissions and immunity is part of the design aim, it does not constitute part of the specification requirement. This is because EMC regulations vary in different parts of the world, and so the level of EMC performance required depends on where a manufacturer wishes to sell his equipment. Adopting the lowest or highest common denominator has never been acceptable. A further difficulty is that it is notoriously difficult to get consistent results from emissions testing.

## 5.3 Specifications in development

It is now believed that the two original specifications are not sufficient to meet all the needs of ATM over copper, and there are two other areas of active development within the ATM Forum as a result.

### 5.3.1 25.6 Mbit/s Desktop Alliance proposal

Several companies were not happy with the decision regarding the line code and bit rate for the mid-range PHY. As a result a body known as the Desktop ATM 25 Alliance [7] was formed with a view to developing an alternative.

The rationale for this development was that there was now a need for ATM delivery to PCs, rather than the high-end workstations at which the original mid-range PHY was aimed. This need was seen as urgent to prevent users being seduced by other technologies (such as Fast Ethernet), and very cost sensitive. The argument was that the mid-range PHY was too costly, complex and late to meet this newly identified market.

In order to meet the aims of simplicity and quick time to market, the development was based on existing technology — Token Ring devices were used as the basis for the system. This explains the unusual 25.6 Mbit/s line rate. A deliberate decision not to use SONET framing had been taken (to decrease the system complexity), but a SONET rate could still have been used. This system, however, uses 4B5B block coding — this allows the use of control characters which, for example, avoids the need for HEC cell delineation. 4B5B encoding with a standard 32 MHz Token Ring clock chip gives the 25.6 Mbit/s data rate.

Although this proposal had originally been developed outside the ATM Forum, the importance of recognition by the Forum was accepted by the Desktop Alliance, and so the specification was proposed to the Forum. There was initially some resistance, largely on a matter of principle. There had already been a very long and passionate debate before the agreement to have just a single mid-range PHY. One of the prime purposes of the ATM Forum is to try and prevent market segmentation and thereby produce economies of scale, and reduce customer confusion. However, the issue was reopened, and after another major debate, the principle of a single solution for this particular case was abandoned. The original Alliance proposal has been somewhat modified during its passage through the Forum, and has now gone out for a final ballot.

### 5.3.2 155 Mbit/s over Category 3 UTP

There are two reasons for this development. The main one is the perceived need for a 155 Mbit/s solution for the installed base of category 3. This is at least arguably true — there is no reason why user bandwidth requirement should be a function of current installed infrastructure type. Further, even if the category 3 and category 5 solution have the same bit rate, there is still a case for two different specifications, in order that the category 5 owner does not have to bear an unnecessarily high cost for complex equipment. The other reason is related to the concern over the EMC properties of the NRZ 155 Mbit/s solution. If a solution is developed which can operate over category 3 from the EMC point of view, it would naturally be more than satisfactory over category 5.

The main contender for this role is a 64-CAP variant of the mid-range PHY — at the functional level only the bit rate is different. This has several advantages; it is a very

easy specification to write since it is so similar to the mid-range PHY, it has similarly advantageous spectral properties, and it is not a great deal more complex to implement, in spite of the increased bandwidth.

#### 5.4 Future developments

A variety of products, specifications and proposals exist at rates between 12.96 and 155.52 Mbit/s for ATM over UTP cable. These are summarised in Table 2, which also contains the original 155 Mbit/s specification for STP cable (only) contained in UNI 3.0. This was not discussed in detail as it was only defined for STP cable which is not now regarded as a general-purpose building wiring type, and is not in widespread use.

Table 2 Summary of ATM over copper specifications.

| Bit rate (Mbit/s) | Cable type                  | Reach (m) | Encoding   | Framing | Status               |
|-------------------|-----------------------------|-----------|------------|---------|----------------------|
| 155.52            | STP                         | 100       | NRZ, 8B10B | SONET   | UNI 3.0              |
| 155.52            | UTP5, STP                   | 100       | NRZ        | SONET   | Published            |
| 155.52            | UTP3, STP, 120 $\Omega$ FTP | 100       | 64-CAP     | SONET   | In development       |
| 51.84             | UTP3                        | 100       | 16-CAP     | SONET   | Published            |
| 25.92             | UTP3                        | 170       | 4-CAP      | SONET   | Published            |
| 12.96             | UTP3                        | 200       | 2-CAP      | SONET   | Published            |
| 25.6              | UTP3, STP, 120 $\Omega$ FTP | 100       | NRZ, 4B45  | None    | Out for final ballot |

155 Mbit/s represents the limit of perceived demand at present, rather than the infrastructure or technology limit. Using the techniques of crosstalk and echo cancellation, and multi-pair operation already described, it is possible to achieve 622 Mbit/s over 100 m of UTP; demonstration systems already exist, although EMC issues have not been fully explored. However, 622 Mbit/s links would be used for switch-to-switch rather than hub-to-workstation connections, in which case 100 m reach may not be adequate and a longer reach fibre solution would be necessary. The current development of a 622 Mbit/s UNI specifically excluded a copper-based solution for this reason.

## 6. Residential ATM

In spite of the massive development effort, fully functional ATM networks consisting of standardised components are still a long way off. The technology is very immature and equipment which does exist tends to be at least partly proprietary and very expensive (although significant price decreases have already been seen). It might

therefore seem premature at best and possibly extremely perverse to even think about ATM in the residential environment where low cost is of paramount importance. However, a reconsideration of the essential purpose and benefits of ATM might alter this view.

ATM is mainly concerned with the efficient multiplexing, switching and delivery of an arbitrary mixture of asynchronous and isochronous (time-bounded) services, with a particular emphasis on the latter. While narrowband isochronous services (e.g. voice) are obviously of vital importance in the business environment, the case for broadband isochronous services is less clear. There is increasing interest in video distribution in certain sectors, e.g. financial news broadcast, but this is not a well-developed area.

The reverse is true in the domestic environment; although data and information are now regarded as significant here, the main driver behind residential broadband is the need to deliver video-based entertainment services. Further, it is now thought important to be able to deliver a simultaneous mixture of services, i.e. either multiple entertainment channels or information plus entertainment, which will be used concurrently by different family members in different parts of the home. This requires that the multiplexing capability and hence ATM delivery is maintained at least into the home, rather than the ATM being terminated at some point back in the network. The question then becomes: 'Should the distribution mechanism within the home also be ATM based, or should the different services be split out and distributed in some other way?'

### 6.1 In-home ATM distribution

There are several advantages in maintaining ATM delivery right into the various items of CPE:

- it avoids the cost and complexity of a multiservice hub,
- it allows a single interface type, with cost and customer acceptance benefits,
- it allows flexible bandwidth delivery and hence flexible and upgradable services,
- it permits a mixture of traffic types into a single item of CPE, e.g. as would be required for a multimedia PC.

However, the overriding consideration remains cost — this is now the consumer electronics market, so how much cost and complexity does an ATM interface add to CPE? There are two aspects to this:

- the cost of having a higher bandwidth interface than is strictly required for a particular service,

- the cost of the ATM specific functionality such as segmentation and reassembly, (HEC) cell delineation, framing (if used), etc.

These costs have not yet been quantified, but it should be noted that the higher bandwidth disadvantage is offset by the advantage of a single interface type, and any packet-based system will have to have some equivalent of the ATM specific functionality.

## 6.2 Non-ATM in-home distribution

In order to decide whether ATM-based distribution is sensible, it is necessary to consider the alternatives. These fall into two classes:

- a multiservice hub with a variety of predefined standard interfaces,
- a non-ATM general-purpose distribution system.

The multiservice hub is likely to be a complicated, expensive and inflexible. There will need to be physical layer, protocol and signalling conversion between ATM and a variety of other systems such as RS232, G703, Ethernet, etc. Finding suitable interfaces for the high bandwidth video services is likely to be difficult, but deciding on the mixture of interfaces is likely to be even harder. There is the danger of over-provisioning — making people pay for interfaces they do not need. A modular, plug-in approach might seem attractive but although this method is common in the business environment, and has arguably already been accepted in the domestic case, i.e. expansion slots in PCs, there is still a great reluctance among manufacturers to do this because there is still a pre-provisioning cost, and there are damage and liability issues.

There are non-ATM distribution systems. Most of the LAN-based ones lack suitable isochronous support. Others such as P1394 [8] and CEBus have been at least partly targeted at the domestic environment, but these have fundamental inadequacies (reach and bandwidth respectively) when considered as general-purpose, broadband home distribution systems. A fuller debate on the alternatives can be found elsewhere [9].

So while there is still a cost issue regarding incorporating ATM interfaces in CPE, the alternatives do not look terribly attractive and it is at least worth investigating in greater detail all the technical and cost issues involved. This is the subject of much current work.

## 6.3 ATM in the residential environment

Accepting that there is a case for ATM in the home then two questions arise. The rationale behind ATM over copper in the business market was the existence of a large installed base of a well-defined and relatively high-performance (UTP) cable. No comparable installed base exists in the domestic environment, especially in the UK, and so the issue of whether copper in general and UTP copper in particular is the most suitable transmission medium for this market needs to be revisited. If it turns out that copper is the preferred medium, then the question is whether any of the systems already developed for the business market are also likely to be suitable for the home market.

In order to consider the home distribution option and technologies, it is first necessary to derive some functional requirements. There is as yet no generally agreed set of requirements, but a consensus is gradually emerging among the various bodies working on this matter which can be summarised as follows:

- simultaneous delivery of multiple services to multiple separate locations,
- support for an arbitrary mix of asynchronous and isochronous services with a well-defined QOS,
- network delivered services likely to have asymmetric bandwidth requirements,
- CPE-to-CPE in addition to network-to-CPE communication highly desirable (this leads to a symmetric CPE interface requirement),
- very low cost interfaces,
- robust, self-configuring network,
- low cost/complexity entry level systems with a well-defined upgrade path.

Where there is less agreement is on the precise parameters associated with these requirements, in particular the bandwidth needed. This depends on one's view of the services, delivery mechanism and topology (e.g. shared or switched media). However, an increasingly quoted figure is for a distribution bandwidth of least 25 Mbit/s, preferably full duplex. This allows several simultaneous (compressed) video channels with some provision for data.

## 6.4 Distribution options

Given the lack of an installed base, it is sensible to consider all the distribution technologies. This is particularly important as the cost of installing a new physical infrastructure may be considerable and there could be consumer resistance, in which case non-physical



transmission or the reuse of existing infrastructure might be preferable as long as the functional requirements can be met.

#### 6.4.1 Radio

Various radio distribution systems exist, developed for the wireless LAN market. However, when compared with the above list of functional requirements they are deficient in three areas — cost, bandwidth and isochronous support. The cost is partly a function of the immaturity and low volume of the technology. It is also fair to discount the cost of not having to install an infrastructure; however, this is taking a system view of the situation, whereas the CPE manufacturer is likely to take a much narrower view of the impact on his particular box. Bandwidth is a real problem, which partly arises from limited spectrum availability; current systems have a maximum bandwidth of about 1 Mbit/s which is quite inadequate, and of course the nature of the medium means that this is broadcast, shared bandwidth. The development of Hiperlan [10] was thought to offer the prospect of broadband isochronous radio distribution, but this only has a (shared, half-duplex) bandwidth of about 12 Mbit/s, and the isochronous support is fairly rudimentary and unlikely to offer a sufficient quality of service.

While radio avoids having to install a wired infrastructure, it is not without its own problems — for example, guaranteeing coverage in a complicated home layout (as opposed to an open plan office) and preventing performance-affecting interference in multiple occupancy buildings (flats).

#### 6.4.2 Infra-red

Diffuse infra-red systems tend to be cheaper than radio ones, but apart from this suffer the same disadvantages of low bandwidth (~ 1 Mbit/s) and poor isochronous support. They have an additional disadvantage in that coverage is limited to a single room and so infra-red transmitters would have to be connected to a wired backbone to provide multi-room coverage.

#### 6.4.3 Existing copper

Using some existing installed infrastructure type offers another means of preventing the expensive and disruptive installation of new cabling. There are three common types of infrastructure which could be used:

- the AC mains power distribution system,
- existing telephone wiring,
- cable TV coaxial systems.

Mains wiring is the only one of these which is ubiquitous in the sense of being present in all homes, and

going to all rooms within a home. Systems do exist for transmitting data over mains wiring — these use spread spectrum techniques to overcome the severe noise problems. They tend to be very low bandwidth, again it is a shared media environment, and there are obvious safety concerns as well as the security issue of signals not being constrained within a single house.

Existing telephone wiring is sub-category 3 standard and is not very well controlled from a topological point of view, i.e. there can be unterminated spurs of cable which will cause reflections. It is possible to devise robust transmission systems which can cope with that kind of environment, but this involves complexity and hence expense. Telephone wiring is not ubiquitous within most houses, and tends not to go to the places where the new services would need to be delivered.

A significant proportion of homes now have a coaxial cable distribution system for cable TV. This has a passive tree and branch topology. This could be used for a broadband broadcast distribution system, which could co-exist with the analogue TV signals. The problem comes with the upstream channel. This is a passive, shared bandwidth system and there will be contention for the upstream channel not only from multiple devices within the home, but from other homes on the same cable. This gives rise to a low upstream bandwidth, effectively prevents intra-house communication, and, as with the power line case, there are security issues with the signals not being constrained to a single house.

#### 6.4.4 Fibre-optic cable

It therefore seems that no system based on non-physical medium transmission, or an existing infrastructure type can provide anything approaching the required functionality, and some new type of infrastructure will need to be installed. The advantage of a new infrastructure is that it can be specified and controlled to a certain degree. This means that it is not necessary to take a lowest common denominator approach to the transmission system which can be cheaper and simpler as a result. However, the new cabling need not necessarily be copper based. It is recognised that the domestic environment is fairly hostile from the EMC immunity point of view and so fibre optics should at least be considered. There would seem to be insuperable installation difficulties for conventional fibre. It will not be possible to control and restrict bend radii, or ensure adequate connectorisation. However, plastic fibre is rather more robust and amenable to low-cost, non-expert installation. There are still issues to do with bending, and laser safety, however, and the current cost of optoelectronics. This is an interesting possibility which is being investigated further.

### 6.5 Detailed requirements and constraints

The above discussion leads to the conclusion that it is going to be necessary to install some new distribution system, probably based on UTP copper in the absence of any suitable alternative. Given this working assumption, more specific requirements of the distribution system can be derived, and any constraints peculiar to this environment considered.

Studies have shown that retrofitting cabling which therefore involves tortuous routeing (round doors, walls, etc) leads to a fairly high reach figure — 50 m between active nodes is a commonly used figure. While this may sound generous, it must be remembered that this is an uncontrolled, unplanned environment and any arbitrary routeing must have a high probability of being less than the maximum reach.

Low cost is fundamental which means that no non-essential functionality should be included.

Class B emissions, level 2 (3 V/m) immunity is a legal EMC requirement in the residential environment, but recent studies have shown that this may not be sufficient to be immune from nearby amateur radio transmitters, operating within legal power limits; 10 V/m immunity may be required. The situation would be eased by the use of the better balanced category 5 rather than category 3 UTP.

### 6.6 Suitability of current business specifications

The specifications listed in Table 2 can be considered in the light of the general and specific requirements. There are two obvious candidates:

- the mid-range PHY operating at 51 or 25 Mbit/s,
- the 25.6 Mbit/s system.

While the mid-range PHY has the required bit rate and range, and good EMC properties, it is, however, a fairly complex system and may be too expensive, especially with the SONET framing which is probably unnecessary in the domestic environment. Chips have also been slow to appear.

The 25.6 Mbit/s system is simpler, has no framing and so should be cheaper. The bandwidth, however, is right at the bottom of the required range, and there are some concerns over the EMC performance.

Therefore neither is ideal, which is hardly surprising since neither was developed for this purpose. The requirements are not as stringent as for the business environment in some ways; the reach is 50 rather than 100 m and

the use of category 5 cable could be mandated, there is no installed base of Category 3 to cater for. It is therefore conceivable that a new, simple, low-cost PHY targeted at this environment could be produced, and there are certain moves in this direction. However, whether this actually happens will be decided on political and time scale grounds, as much as technical issues.

## 7. Conclusions

The ability to deliver ATM over copper in the business environment is essential if ATM is to fulfil its promise of becoming the technology which provides a fully integrated communications system in the sense of both delivering a multiplicity of service types, and removing the LAN/WAN distinction. At the physical layer, ATM over copper is essentially no different to any other high-speed transmission system, and there is proof that this can be achieved by the existence of such systems as CDDI. There are several ATM Forum specifications at an advanced stage which provide for a variety of transmission rates between 12.96 and 155.52 Mbit/s. Systems running at 622 Mbit/s over 100 m of category 5 UTP have been demonstrated, but are not currently being specified due to perceived lack of commercial demand.

Residential ATM distribution is at a less advanced stage. There are a lot of arguments in its favour and a lot of recent interest in its development. However, it is essential to demonstrate that it can be done at a cost which is acceptable to CPE manufacturers. This fact, plus the constraints of the residential environment, may mean that the development of a simple new PHY specifically for this environment is necessary.

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## Appendix

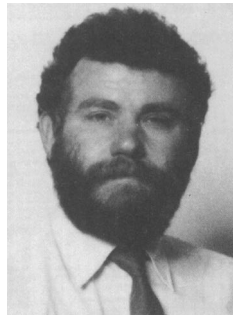
### List of acronyms

|       |                                                         |
|-------|---------------------------------------------------------|
| ANSI  | American National Standards Institute                   |
| n-CAP | carrierless amplitude phase with n constellation points |
| BPRIV | binary partial response 4                               |
| CDDI  | copper distributed data interface                       |
| CPE   | customer premises equipment                             |
| DSP   | digital signal processing                               |
| EIA   | Electrical Industries Association                       |
| EMC   | electromagnetic compatibility                           |
| FDDI  | fibre distributed data interface                        |

|       |                                      |
|-------|--------------------------------------|
| FTP   | foil screened twisted pair cable     |
| HEC   | header error check                   |
| ISO   | International Standards Organisation |
| LAN   | local area network                   |
| MLT-3 | multi-level transmission 3 (levels)  |
| NRZ   | non-return to zero                   |
| PC    | personal computer                    |
| PHY   | physical layer                       |
| QAM   | quadrature amplitude modulation      |
| QOS   | quality of service                   |
| SDH   | synchronous digital hierarchy        |
| SONET | synchronous optical network          |
| STP   | screened twisted pair                |
| UNI   | user network interface               |
| UTP   | unscreened twisted pair              |
| WAN   | wide area network                    |

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More recently the same team has become involved in the development and specification of systems for broadband residential distribution.